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Day 3

Run-time Optimization of Finite Difference Methods

Spring 2025

PH 367: Computational Physics Lab III

Heat equation in our code

```
while t < tf:
    rate = np.zeros(N)
    for i in range(1, N-1):
        rate[i] = alpha * (T[i-1] - 2 * T[i] + T[i+1]) / dx**2
    dT = rate * dt
    T += dT
    t += dt
```

Calculating dTdt one chunk of wire at a time

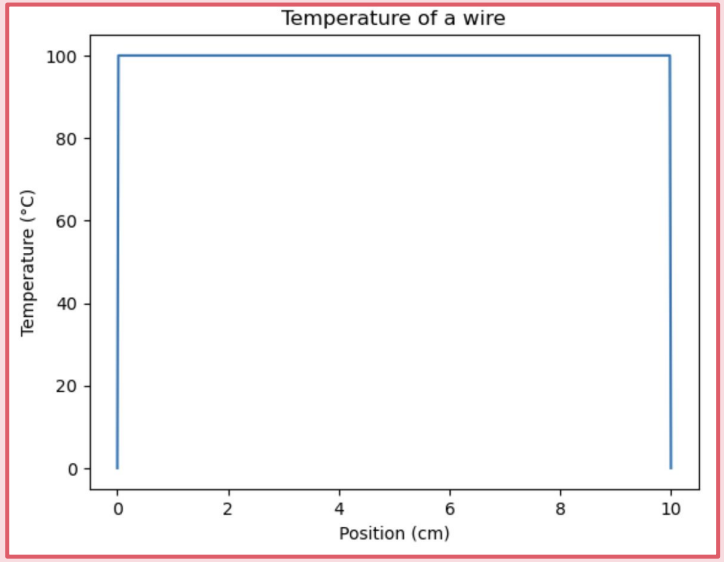
$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

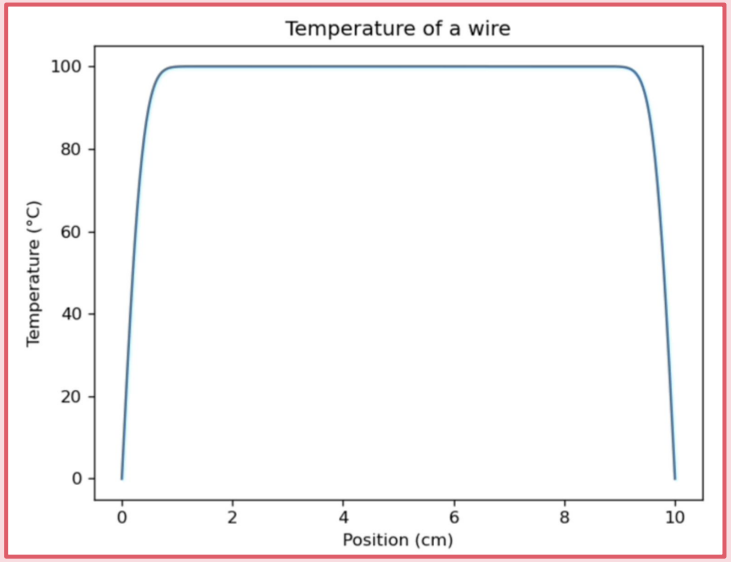
Same code using functions

```
def d2Tdx2(T, i, dx):
    return (T[i-1] - 2 * T[i] + T[i+1]) / dx**2

def dTdt(T, N, dx, alpha=1.27e-4):
    rate = np.zeros(N)
    for i in range(1, N-1):
        rate[i] = alpha * d2Tdx2(T, i, dx)
    return rate

while t < tf:
    dT = dTdt(T, N, dx) * dt
    T += dT
    t += dt
```

New initial conditions

Wire starts at 100 degrees, both ends held to 0 degrees





Extending model further in time

```python
while np.max(T) > 1:

    t0 = time.time()

    rate = np.zeros(N)

    t1 = time.time()

    for i in range(1, N-1):
        rate[i] = alpha * d2Tdx2(T, i, dx)

    t2 = time.time()

    T += rate * dt
    t += dt

    t3 = time.time()

    time_zeros += (t1 - t0)
    time_calc += (t2 - t1)
    time_update += (t3 - t2)

print("Time to initialize rate array:", time_zeros)
print("Time to calculate dTdt:", time_calc)
print("Time to update T and t:", time_update)
```

```
Time to initialize rate array: 0.4824798107147217
Time to calculate dTdt: 77.13064956665039
Time to update T and t: 0.5248582363128662
```

Current model takes a lot of time to run...

From `t0` to `t1`, initializing the rate array, 0.5 seconds total

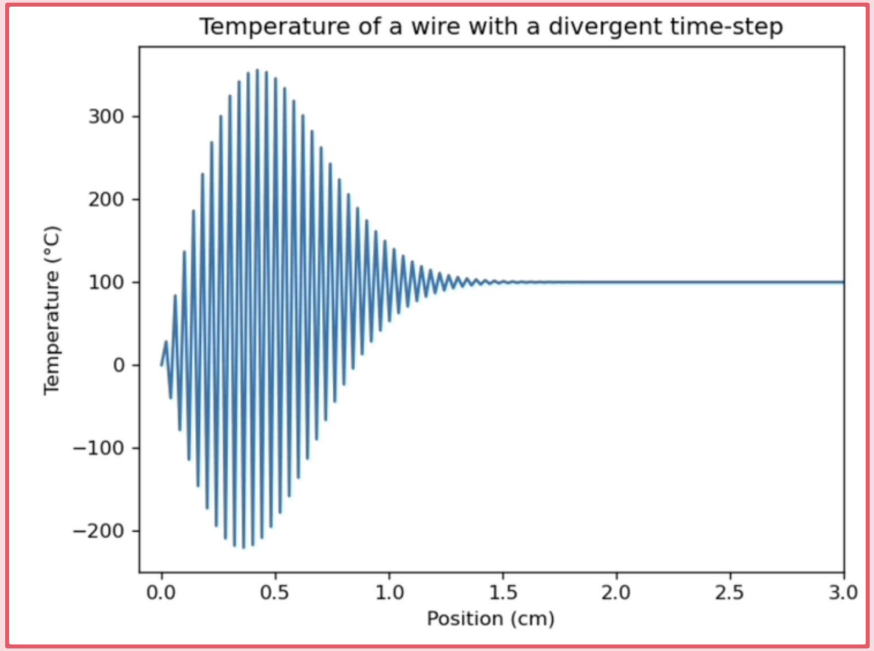
From `t1` to `t2`, calculating `dTdt`, 77 seconds total!!!

From `t2` to `t3`, updating `T` and `t`, 0.5 seconds total

The runtime culprit!

```
def d2Tdx2(T, i, dx):
    return (T[i-1] - 2 * T[i] + T[i+1]) / dx**2

def dTdt(T, N, dx, alpha=1.27e-4):
    rate = np.zeros(N)
    for i in range(1, N-1):
        rate[i] = alpha * d2Tdx2(T, i, dx)
    return rate

while t < tf:
    dT = dTdt(T, N, dx) * dt
    T += dT
    t += dt
```

What if we just increase the time step?



Maybe let's keep `dt = 1` for now...

Today’s activity

1. Change the calculation of `dTdt` to improve the runtime
2. Explore how larger time-steps cause divergent behavior